

Copper Wire Insertion in Serpentine Channel:

Symmetry Breaking and Rabi-like Mode Splitting in a Honeycomb-Serpentine Polarization Interferometer

DIY Optics Lab — Busan Medical Center | Experimental note, June 07, 2026 | GitHub archive:
honeycomb-serpentine-series

Abstract. We report quantitative optical differences observed when a copper wire is threaded through the serpentine groove of a honeycomb-serpentine Moire polarization interferometer driven by an N20 DC micro-motor. Frame-by-frame OpenCV analysis of five video takes (with wire) versus four takes (without wire) reveals: (i) a 43% increase in mean brightness, (ii) a 56% reduction in Michelson fringe contrast, (iii) the emergence of a new FFT mode ($\omega_3 \sim 0.067$ normalized), (iv) a dramatic drop in time-reversal symmetry score (0.639 to 0.116), and (v) a 33% reduction in detected dot count. We interpret these observations in terms of symmetry breaking, cavity-mediated mode coupling, and an analogy with Rabi-like anti-crossing. Notably, the behavior is qualitatively opposite to what would be expected from graphene-based conductive overlays, pointing to a bulk-conductivity versus surface-conductivity distinction.

1. Experimental Setup

The interferometer consists of a honeycomb-patterned polarizing film (input privacy film, ON/OFF selectable) stacked with a serpentine-patterned polarizing film, mounted on a book platform and illuminated by ambient / reflected light. An N20 DC micro-motor (AA battery, ~ 1.5 V) rotates the honeycomb layer, generating dynamic Moire fringes. Circuit parameters are monitored with an ASAHI 4201 digital multimeter.

Copper-wire condition (WITH):

- A thin copper wire is inserted lengthwise into the serpentine channel groove.
- Wire acts simultaneously as a conductive element and an optical waveguide boundary.
- 5 video takes recorded: 20260607_171753 through 20260607_171950.

Baseline condition (NO wire):

- Serpentine channel is bare (no wire insert).
- 4 video takes recorded: 20260607_174627 through 20260607_174750.

All videos: 1280 x 720 px, ~ 30 fps, compressed with FFmpeg (CRF 28, H.264) and archived on GitHub. OpenCV-based analysis extracts per-frame brightness and contrast from the central 50% ROI; blob detection counts discrete bright domains (dots).

2. Quantitative Results

2.1 Core photometric metrics

Metric	WITH wire	NO wire	Ratio (WITH / NO)
Mean brightness (AU)	126.7 ± 3.5	88.4 ± 6.2	1.43x
Michelson fringe contrast	0.063 ± 0.002	0.143 ± 0.036	0.44x
Brightness std (temporal)	3.54	6.19	0.57x

Modulation depth	0.040	0.097	0.41x
Time-reversal symmetry score	0.116	0.639	0.18x
Dot count (mean ± std)	11.6 ± 3.8	17.4 ± 4.0	0.67x
Spatial spread (px)	113.5	105.3	1.08x

Table 1. Summary of OpenCV frame analysis. Each value for WITH wire is the mean across 5 takes; NO wire across 4 takes. Ratio < 1 indicates suppression by copper wire.

2.2 FFT dominant frequency modes

Mode	Norm. freq.	Amplitude (WITH)	Amplitude (NO)	Interpretation
omega-1	0.0083	276	505	Fundamental motor rotation
omega-2	0.042	47	55	2nd harmonic / Moire beat
omega-3	0.067	29	---	New mode: apparent only WITH wire

Table 2. FFT peak amplitudes (brightness time-series, 120 sampled frames). The appearance of omega-3 exclusively in the WITH-wire condition is the central spectral signature.

3. Symmetry Analysis

3.1 Time-reversal (T) symmetry

The time-reversal symmetry score is computed as the Pearson correlation between the first and time-reversed second half of the brightness time-series. A score near +1 indicates a system whose dynamics are unchanged under $t \rightarrow -t$; a score near 0 indicates broken T-symmetry.

Condition	T-symmetry score	Interpretation
WITHOUT wire	0.639	Near-symmetric: reversible oscillatory dynamics
WITH wire	0.116	Symmetry broken: irreversible dissipative coupling

Table 3. Time-reversal symmetry scores.

The copper wire introduces a dissipative (resistive) channel that breaks T-symmetry. In the language of the effective axion-phase framework, this corresponds to a spatially inhomogeneous $\theta(r)$ field acquiring a nonzero imaginary part, i.e., the system transitions from a Hermitian to a non-Hermitian Hamiltonian regime.

3.2 Spatial symmetry

The 8% increase in spatial dot spread (113.5 vs 105.3 px) combined with a 33% reduction in dot count suggests that the copper wire selectively extinguishes certain spatial modes while expanding the coherence radius of the surviving modes. This is consistent with a translational symmetry reduction: the periodic serpentine lattice acquires a new periodicity imposed by the wire, lowering the effective spatial symmetry group.

4. Rabi-like Mode Splitting — Physical Interpretation

4.1 Anti-crossing phenomenology

In cavity quantum electrodynamics, Rabi splitting occurs when two resonant modes couple with strength g , producing dressed-state eigenfrequencies:

$$\omega_{\pm} = (\omega_1 + \omega_2) / 2 \pm \sqrt{g^2 + (\Delta)^2 / 4}$$

where $\Delta = \omega_2 - \omega_1$ is the bare detuning and g is the coupling constant. The emergence of ω_3 in the WITH-wire condition is interpreted as the lower polariton branch of such a coupling: the copper wire mediates coherent energy exchange between the optical Moiré mode and the mechanical/electromagnetic mode of the wire itself. Verification of full anti-crossing requires a voltage sweep (DC bias via the wire) to tune Δ continuously and observe the avoided crossing.

4.2 Coupling strength estimate

From the FFT data, the frequency splitting between ω_2 and ω_3 is $\Delta\omega \sim 0.025$ (normalized). If this reflects $2g$ at near-resonance ($\Delta \sim 0$), then $g \sim 0.013$. The fundamental linewidth $\kappa \sim 0.008$ (estimated from peak width), giving $g/\kappa \sim 1.6$, which is consistent with the onset of coherent coupling ($g > \kappa/2$). This is a preliminary estimate; a DC-sweep experiment is needed to confirm the anti-crossing topology.

4.3 Why the image looks sharper despite lower contrast

The Michelson contrast is paradoxically lower WITH the wire (0.063 vs 0.143), yet the image subjectively appears sharper. This is explained by the distinction between fringe contrast (amplitude of intensity oscillation) and spatial coherence (phase locking across the field). The copper wire suppresses noisy higher-order modes (dot count -33%), leaving a smaller number of spatially coherent, phase-locked modes. Human visual perception of sharpness is dominated by spatial coherence and edge definition rather than raw intensity modulation depth.

5. Contrast with Graphene — Bulk vs. Surface Conductivity

Key insight: The copper wire behaves in a manner qualitatively opposite to what a graphene overlay would produce, illuminating a fundamental difference between bulk conductors and atomically thin conductors in Moiré optical systems.

Property	Copper wire (this work)	Graphene (expected)
Conduction type	Bulk 3D metallic	2D surface / Dirac fermion
Optical transmission	Opaque (reflective)	~97.7% transparent
Effect on fringe contrast	Suppresses (0.063)	Modulates weakly, preserves
T-symmetry	Strongly broken (0.116)	Weakly broken or preserved
New FFT mode	Yes (ω_3 emerges)	Unlikely; no bulk resonance
Dot count	Reduced -33%	Minimal change expected
$\theta(r)$ field effect	Bulk dielectric screening	Topological surface term
Coupling channel	EM + mechanical (wire mass)	EM only (massless carriers)

Table 4. Predicted comparison between copper wire (experimental) and graphene overlay (theoretical expectation based on published graphene optics literature).

In graphene, the optical conductivity is frequency-independent in the visible range ($\sigma = e^2/4\hbar$), giving 2.3% absorption per layer and negligible mechanical coupling. Graphene would therefore act as a weak, uniform attenuator that preserves fringe contrast and does not break T-symmetry appreciably. The copper wire, by contrast, introduces three qualitatively new effects absent in graphene: (1) specular reflection that redirects optical modes, (2) a mechanical resonance of the wire itself that couples to the Moiré dynamics, and (3) Ohmic dissipation that irreversibly breaks T-symmetry. This trifecta produces the new ω_3 mode and the sharpness enhancement with reduced Michelson contrast.

6. Connection to Effective Axion Phase Field $\theta(r)$

The honeycomb-serpentine interferometer has been theoretically framed as a system supporting a spatially alternating effective $\theta(r)$ phase field, in analogy with axion electrodynamics in topological condensed matter. The copper wire experiments provide the first experimental perturbation of this field:

- The wire locally pins the $\theta(r) = 0$ boundary condition along its length (conducting surface forces the tangential E-field to zero), creating a hard topological defect line in the otherwise smoothly varying phase field.
- This defect breaks both translational and time-reversal symmetry, consistent with the observed T-symmetry score collapse from 0.639 to 0.116.
- The new FFT mode ω_3 may represent a localized mode bound to the defect line, analogous to topological edge states in Chern-insulator systems.
- Future DC sweep experiments (using the wire as an electrode) will allow independent tuning of the electrical component of $\theta(r)$, decoupling it from the optical component controlled by polarizer rotation.

Dual-variable scheme: Optical θ modulated by motor rotation angle ϕ ; electrical θ modulated by DC bias V through the copper wire. This realizes the independent (ϕ, V) parameter space needed for the full dual-variable θ system proposed in the experimental protocol document.

7. Proposed Next Experiments

Priority	Experiment	Expected observable
1 (High)	DC voltage sweep 0-5V through wire	$g(V)$ curve; anti-crossing if ω_3 tracks V
2 (High)	Resistance verification (4-wire)	Confirm Ohmic contact; rule out capacitive coupling
3 (Med)	Wire diameter series (0.1-1 mm Cu)	Mode splitting vs. wire cross-section
4 (Med)	Replace Cu with resistive wire	Isolate mechanical vs. electrical contribution
5 (Low)	Graphene flake overlay test	Confirm predicted minimal T-symmetry breaking

Table 5. Proposed follow-up experiments.

8. Summary

- Copper wire insertion into the serpentine channel of the honeycomb-serpentine polarization interferometer produces a qualitatively distinct optical state compared to the bare channel baseline.
- The primary signatures are: higher mean brightness (+43%), lower Michelson contrast (-56%), new FFT mode ω_3 , and dramatic T-symmetry breaking (score: 0.116 vs 0.639).
- The subjectively sharper image with the wire present is attributed to spatial coherence enhancement (mode locking) rather than contrast increase, consistent with selective noisy-mode suppression (dot count -33%).
- The observations are interpreted as Rabi-like anti-crossing between the Moire optical mode and a wire-mediated dissipative channel, with preliminary coupling ratio $g/\kappa \sim 1.6$.
- The behavior is predicted to be qualitatively opposite to a graphene overlay, distinguishing bulk-conductor (3D) from surface-conductor (2D Dirac) coupling physics.
- The copper wire realizes the electrical component of the dual-variable $\theta(r)$ system, enabling independent optical and electrical control of the effective axion phase field.

All raw video data, compressed archives, and OpenCV analysis scripts are available at:
[github.com/\[repository\]/honeycomb-serpentine-series](https://github.com/[repository]/honeycomb-serpentine-series)

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